



**Faculty of Mechanical
Engineering and Robotics**

**Department of Robotics and
Mechatronics**



Signal processing and identification in control of mechatronic devices

Topic: Image binarization and morphological operations

Goal: Students are introduced to methods of image binarization. Morphological operations application in binary and grayscale image processing is discussed.

Issues:

1. Grayscale-based image binarization (thresholding),
2. Image morphological operations on binary images,
3. Image morphological operations on grayscale images.

Prowadzący: dr inż. Krzysztof Holak

1. Image tresholding

Image tresholding (binarization) changes grayscale image into binary image. The most common type of binarization is based on the threshold value. For example, all pixels of a value less than threshold become 0, and all pixels of value larger than threshold become 1. Tresholding is one of the first steps of image analysis pipeline. It is used to separate objects from a background in cases when there is large difference in value between object and background pixels.

Tresholding in MATLAB may be carried out by the following operations:

```
threshold = graythresh(im);  
im_bw = im2bw(im,threshold);
```

Function **graythresh()** computes threshold of binarization automatically and function **im2bw()** modifies grayscale image into binary image based on the obtained threshold value. Threshold may also be found manually by reading its value from image histogram. In MATLAB, threshold value is in a range (0,1).

Pixels of binary images are logical truth values (1/0). Therefore it is possible to perform logical operation on the images. For example:

- 1) Logical AND operation: `and(im1, im2)`
- 2) Logical OR operation: `or(im1, im2)`
- 3) Logical Negation: `not(im1)`; or by using function `imcomplement()`

It is possible to carry out arithmetic operations on images (both binary and grayscale):

```
imadd(im1, im2), imsubtract(im1, im2), immultiply(im1, im2).
```

Problems 1.

1. Perform tresholding of images in order to extract objects from a background. Use automatic and manual computation of threshold value.
2. Using tresholding and logical/arithmetic operation write programs that:
 - extracts objects of pixel values $I > T$
 - extracts objects of pixel values $I < T$
 - extracts objects of pixel values I in a range (T_1, T_2)
 - extracts objects of pixel values $I < T_1$ or $I > T_2$, where $T_1 < T_2$

In all cases objects consist of white pixels and T is a given threshold value.

3. Image morphological operations on binary images

Morphological operations carry out image set operations. All of them can be applied to binary as well as grayscale images. In general, image morphological operation is performed by sliding a mask, called structuring element, through an entire image and computing, for each mask position, a logical operations between pixels of a mask and the image patch under the mask. Morphological operation can be illustrated the best in the binary image domain. Next, the effect of particular morphological operation on binary images is described,

The first step is choice of structuring element (a mask):

```
se1 = strel('type',size);
```

Where '**type**' is a type of structuring element, and **size** is its size. After choosing a mask on of the following can be carried out.

1. **Erosion** – erosion filter removes noise in a form of objects smaller than the filter mask. It decreases the area of objects and disconnects objects which are connected by a thin bridges.

```
bw_erode = imerode(im_bw,se1);
```

2. **Dilation** – dilation filter removes noise in a form of a holes in objects. It increases the area of objects and connects objects which are close to each other.

```
bw_dilate = imdilate(im_bw,se1);
```

3. **Opening** – consists of consecutive erosion and dilation operations. It removes noise in a form of a small objects, but does not change the area of larger objects.

```
bw_open = imopen(im_bw,se1);
```

4. **Closing** – consist of consecutive dilation and erosion operations. It removes noise in a form of holes in filled objects, but it does not increase the area of objects.

```
bw_close = imclose(im_bw,se1);
```

In MATLAB, there is implemented function **bwmorph()**, which performs all the mentioned operations, but also provides different and more complex morphological processing functions.

Problems 2

1. Write a program which carries out morphological operations. Test the impact of type and shape of structuring elements.
2. How to make an edge detector using only morphological operations? Implement the algorithm.
3. Test the following methods using `bwmorph()` function: *bothat*, *remove*, *skel*, *thicken*, *thin*, *tophat*. What are the applications of the operations?

4. Morphological operations on grayscale images

Morphological operations can be applied on grayscale images. Morphological filter **imerode()** works as min function, it increases the intensity of the image, suppressing the darkest pixels. On the other hand **imdilate()** filter performs the opposite, it suppresses brightest pixels, making entire image darker. Filters **imclose()** i **imopen()** work similar, but the application of them does not change the geometrical and photometrical structure of images so much as the previous ones.

Problem 3

Test morphological filters on the grayscale image. Remove noise from images given by teaching assistant using morphological filtering.